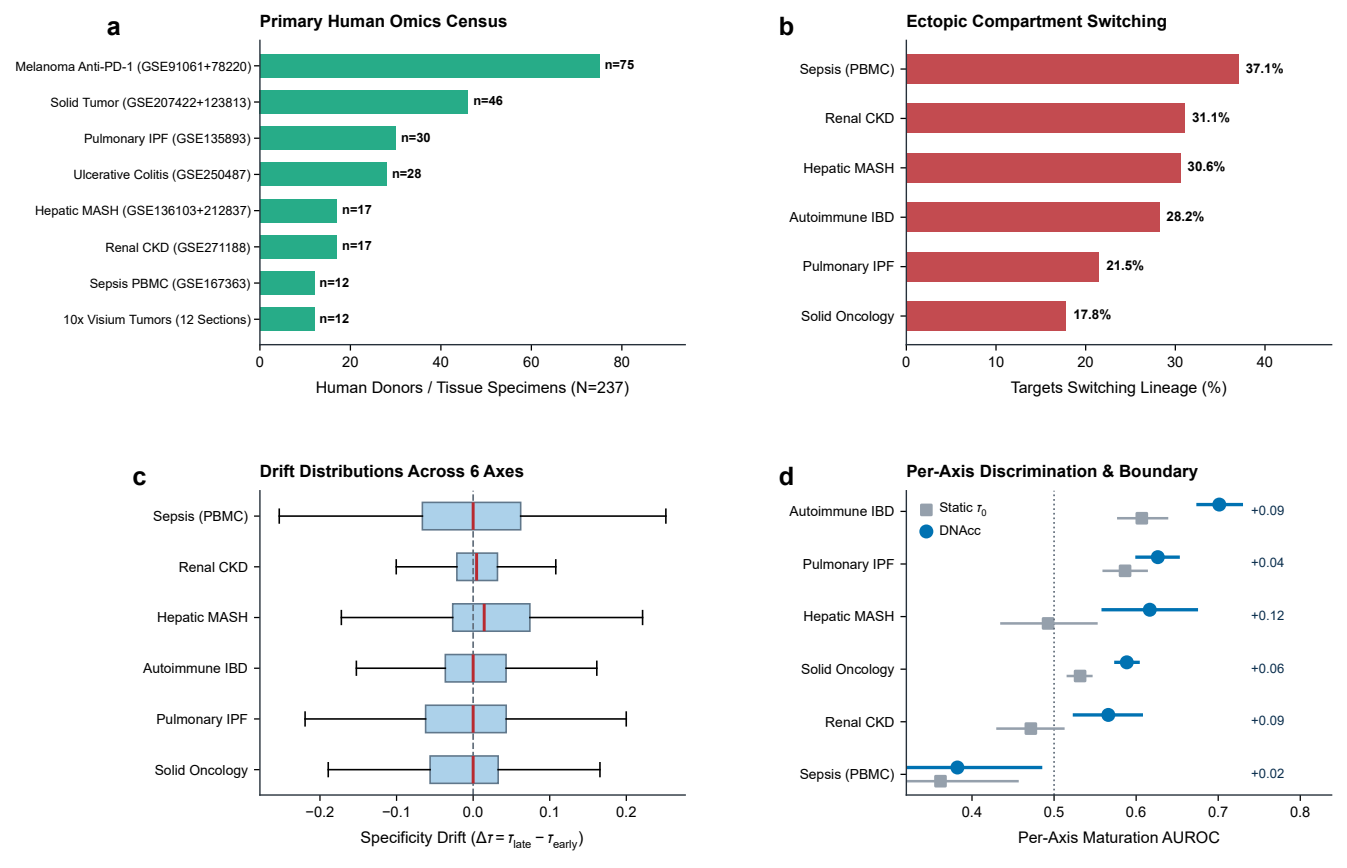

a
Primary Human Omics Census
Melanoma Anti-PD-1 (GSE91061+78220)
Solid Tumor (GSE207422+123813)
Pulmonary IPF (GSE135893)
Ulcerative Colitis (GSE250487)
Hepatic MASH (GSE136103+212837)
Renal CKD (GSE271188)
Sepsis PBMC (GSE167363)
10x Visium Tumors (12 Sections)
n=75
n=46
n=30
n=28
n=17
n=17
n=12
n=12
0
20
40
60
80
Human Donors / Tissue Specimens (N=237)
b
Ectopic Compartment Switching
Sepsis (PBMC)
Renal CKD
Hepatic MASH
Autoimmune IBD
Pulmonary IPF
Solid Oncology
37.1%
31.1%
30.6%
28.2%
21.5%
17.8%
0
10
20
30
40
Targets Switching Lineage (%)
c
Drift Distributions Across 6 Axes
Sepsis (PBMC)
Renal CKD
Hepatic MASH
Autoimmune IBD
Pulmonary IPF
Solid Oncology
−0.2
−0.1
0.0
0.1
0.2
Specificity Drift (Δr = r_late − r_early)
d
Per-Axis Discrimination & Boundary
Static τ0
DNAcc
Autoimmune IBD
Pulmonary IPF
Hepatic MASH
Solid Oncology
Renal CKD
Sepsis (PBMC)
+0.09
+0.04
+0.12
+0.06
+0.09
+0.02
0.4
0.5
0.6
0.7
0.8
Per-Axis Maturation AUROC